SQL QUERIES

1. Creation of Database :-

```
CREATE DATABASE beverages_data;
```

2. Use Table :-

```
USE beverages_data;
```

3. Creation of Table :-

```
CREATE TABLE beverages_data (
   Date DATE,
   Product VARCHAR(100),
   Category VARCHAR(100),
   Company VARCHAR(100),
   Country VARCHAR(100),
   Region VARCHAR(100),
   Unit_Sold INT,
   Price_USD DECIMAL(10,2),
   Revenue_USD DECIMAL(12,2),
   Profit_USD DECIMAL(12,2),
   Customer_Age_Group VARCHAR(50),
   Customer_Gender VARCHAR(10),
   Rating DECIMAL(3,2),
   Hot_Cold VARCHAR(10)
);
```

4. Insertion of Data :-

```
USE beverages_data;

BULK INSERT beverages_data
FROM 'C:\Users\Dell\Desktop\Project\beverages.csv'
WITH (
   FIRSTROW = 2,  -- skip header row
   FIELDTERMINATOR = ',',
   ROWTERMINATOR = '\n',
   TABLOCK
);
```

5. To Verify :-

```
SELECT COUNT (*) FROM beverages_data;
```

DAX QUERIES

Beverages Table

1. Profit Margin (%) :-

```
Profit Margin (%) = DIVIDE(SUM('beverages_data'[Profit]),
                           SUM('beverages_data'[Revenue_USD])) * 100
```

Calendar Table

1. Creation of Table :-

```
CalendarTable = CALENDAR(MIN(beverages_data[Date]), MAX(beverages_data[Date]))
```

2. Day :-

```
Day = DAY('CalendarTable'[Date])
```

3. Day Name :-

```
Day Name = FORMAT('CalendarTable'[Date], "ddd")
```

4. Day Number :-

```
Day Number = WEEKDAY('CalendarTable'[Date], 2)
```

5. Month :-

```
Month = FORMAT('CalendarTable'[Date], "MMMM")
```

6. Month Number :-

```
MonthNumber = MONTH('CalendarTable'[Date])
```

7. Week :-

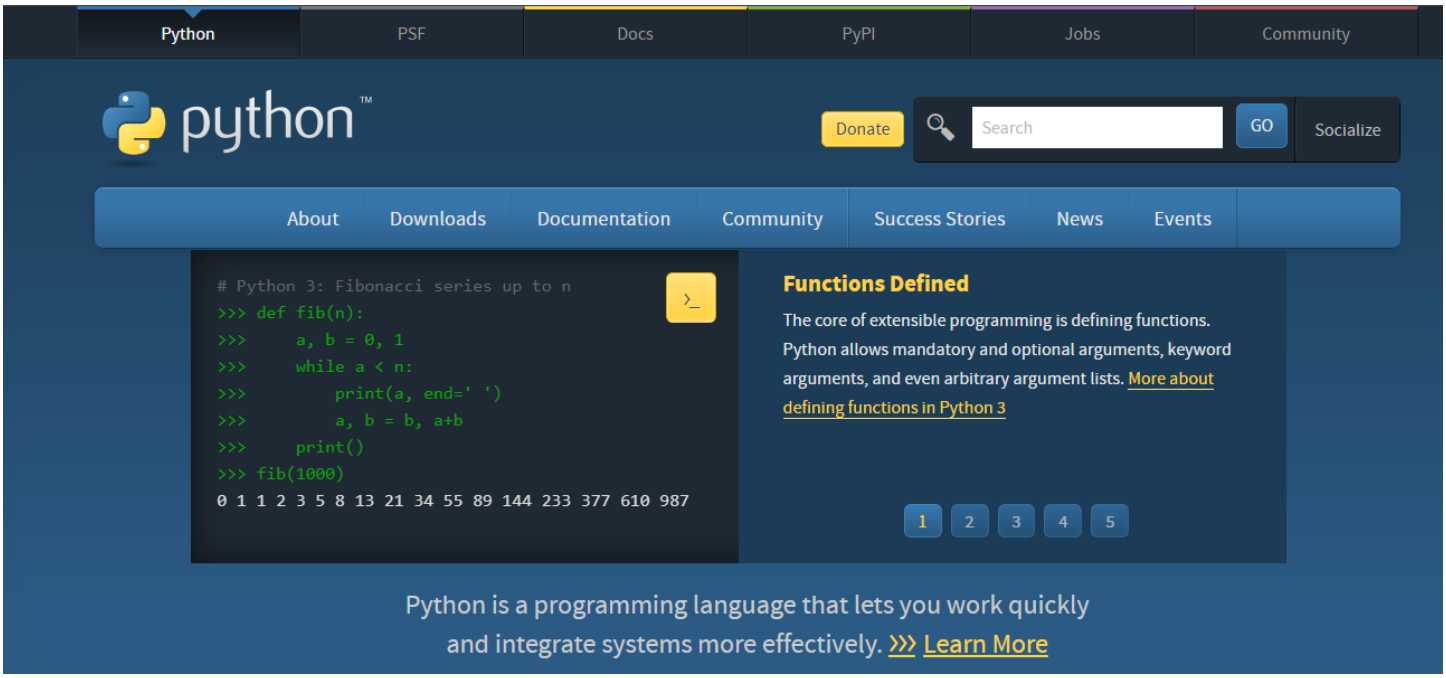
```
Week = WEEKNUM('CalendarTable'[Date],2)
```

8. Year :-

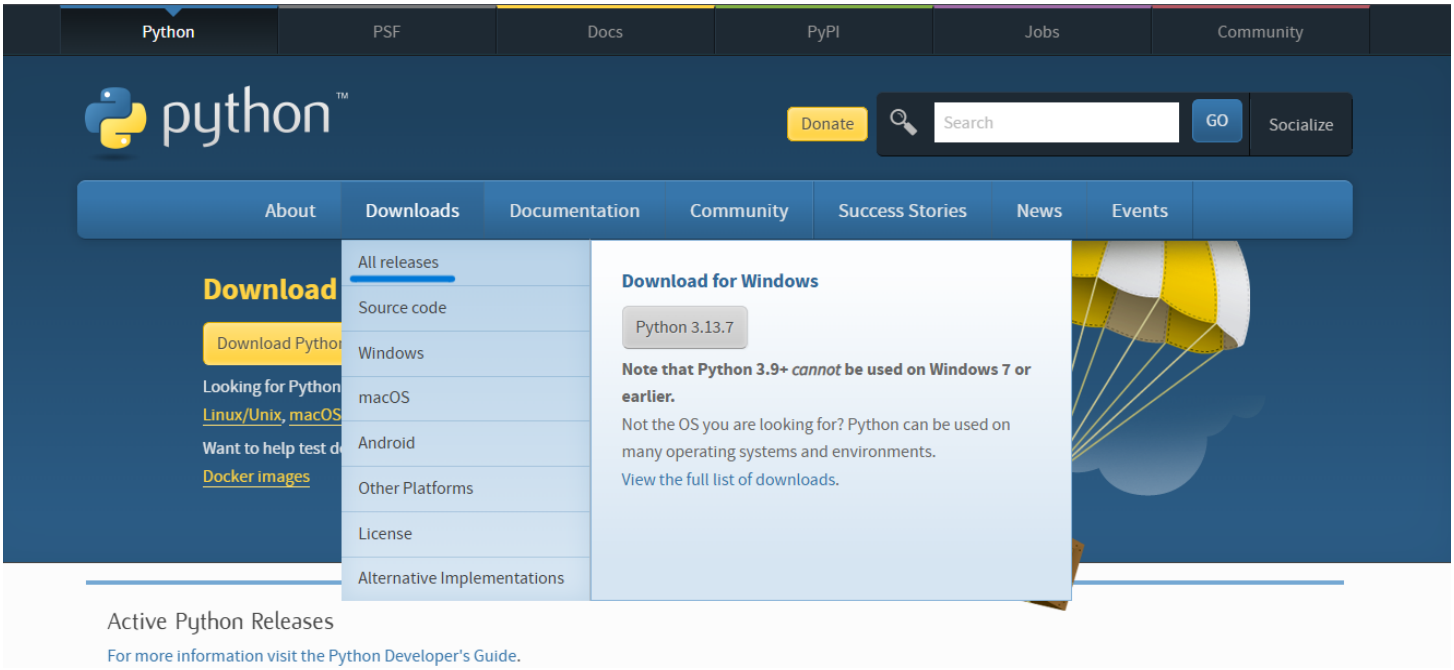
```
Year = YEAR('CalendarTable'[Date])
```

INSTALLATION OF PYTHON AND LIBRARIES

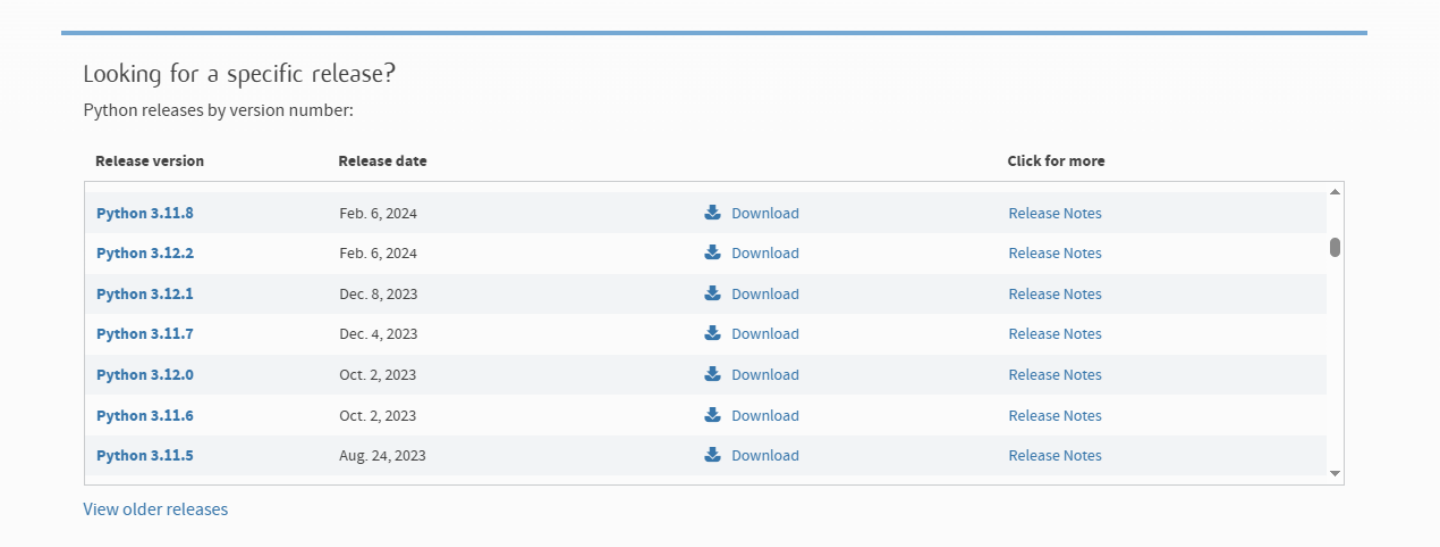
STEP 1 – Open https://www.python.org/ for installation of python version



STEP 2 – Click on DOWNLOADS and then on ALL RELISES



STEP 3 – Now Scroll down for older version of python because libraries like pandas install easily on versions between 3.7-3.12 and then download it .



Looking for a specific release?

Python releases by version number:

Release version	Release date		Click for more
Python 3.11.8	Feb. 6, 2024	Download	Release Notes
Python 3.12.2	Feb. 6, 2024	Download	Release Notes
Python 3.12.1	Dec. 8, 2023	Download	Release Notes
Python 3.11.7	Dec. 4, 2023	Download	Release Notes
Python 3.12.0	Oct. 2, 2023	Download	Release Notes
Python 3.11.6	Oct. 2, 2023	Download	Release Notes
Python 3.11.5	Aug. 24, 2023	Download	Release Notes

View older releases

STEP 4 :- PLEASE visit this YouTube video for better description for the further procedure of installation of python and libraries of python .

https://youtu.be/JB0kIqT1BEk?si=-1D9YwkqYvIYX2Yp

PYTHON SCRIPTS

1. Product and Sum of Rating :-

```
import pandas as pd
import matplotlib.pyplot as plt

# Copy dataset from Power BI
df = dataset.copy()

# Group by Product and calculate average rating
funnel_data = df.groupby('Product').agg({'Rating': 'mean'}).reset_index()

# Sort products by average rating (descending)
funnel_data = funnel_data.sort_values(by='Rating', ascending=False)

# Plotting funnel-style chart (horizontal bar)
plt.figure(figsize=(11, 6))
plt.barh(funnel_data['Product'], funnel_data['Rating'], color='pink')

plt.gca().invert_yaxis()  # Highest rating at the top
plt.xlabel('Average Rating (out of 5)')
plt.grid(axis='x', linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()
```

2. Revenue by Region :-

```
import pandas as pd
import matplotlib.pyplot as plt

# Group by Region and calculate total Revenue
data = dataset.groupby('Region')['Revenue_USD'].sum().reset_index()

# Sort by Revenue in descending order
data = data.sort_values(by='Revenue_USD', ascending=False)

# Funnel Plot
plt.figure(figsize=(8,5 ))
plt.plot(data['Revenue_USD'], range(len(data)), 'o-', color='pink', linewidth=3)
plt.yticks(range(len(data)), data['Region'])
plt.gca().invert_yaxis()  # Invert to show highest at top
plt.xlabel('Revenue (USD)')
plt.grid(axis='x', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

3. Unit Sold by Company :-

```
import pandas as pd
import matplotlib.pyplot as plt

# Group your data
data = dataset.groupby('Company')['Unit_Sold'].sum().reset_index()

plt.figure(figsize=(10, 5))
plt.bar(data['Company'], data['Unit_Sold'], color='pink')

# Change font sizes
plt.xlabel('Company', fontsize=14)      # X-axis label
plt.ylabel('Units Sold', fontsize=14)   # Y-axis label
plt.xticks(rotation=45, fontsize=10)    # X-axis tick labels
plt.yticks(fontsize=10)                 # Y-axis tick labels

plt.tight_layout()
plt.show()
```

4. Profit by Category :-

```
import pandas as pd
import matplotlib.pyplot as plt

# Group by Category and sum up Profit
data = dataset.groupby('Category')['Profit'].sum().reset_index()

# Pie Chart
plt.figure(figsize=(4, 4))
plt.pie(data['Profit'],
        labels=data['Category'],
        autopct='%1.1f%%',
        startangle=90,
        counterclock=False)
plt.title('Profit Distribution by Category')
plt.tight_layout()
plt.show()
```

5. Average Rating by Product :-

```
import matplotlib.pyplot as plt
import pandas as pd

# Ensure numeric and ignore blanks
r = pd.to_numeric(dataset['Rating'], errors='coerce').dropna()
avg = r.mean() if len(r) else float('nan')

fig, ax = plt.subplots(figsize=(2, 1))
ax.axis('off')


ax.text(0.5, 0.48, f"{avg:.1f} / 5", ha='center', fontsize=22, fontweight='bold')
plt.tight_layout()
plt.show()
```

6. Average Profit by Product :-

```
import matplotlib.pyplot as plt

# Calculate average profit
avg_profit = dataset["Profit"].mean()

# Create KPI figure
fig, ax = plt.subplots(figsize=(2,1))
ax.text(0.5, 0.6, f"${avg_profit:,.2f}",
        fontsize=28, fontweight="bold", ha="center", va="center", color="darkgreen")


# Remove axes
ax.axis("off")
plt.tight_layout()
plt.show()
```